

Supplementary Material: Full Reproducibility Configuration

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A. DETAILED HYPERPARAMETER SETTINGS

To ensure full transparency and reproducibility, we list the exact hyperparameter configurations used to generate every figure and table in the main manuscript. All experiments are reproducible using the provided code and configuration files.

TABLE I
SUPPLEMENTARY TABLE S1: FULL REPRODUCIBILITY CONFIGURATION.

EXACT HYPERPARAMETERS USED TO GENERATE REPORTED RESULTS.
DEFAULT LOSS WEIGHTS ARE $\lambda_{\text{MSE}} = 1.0$. THE PERCEPTUAL WEIGHT λ_{perc} IS ADAPTED TO DATASET COMPLEXITY:
0.05 FOR NATURAL SCENES (CIFAR-10) AND 0.10/0.20 FOR SIMPLE STRUCTURES (FASHION-MNIST).

Experiment / Items	Dataset	γ	Batch	LR	λ_{perc}	Noise (Train $\rightarrow$ Eval)
Main @ 5% (Visual / Grad.)	CIFAR-10	0.05	512	1×10^{-3} (Net) 5×10^{-4} (Phys)	0.05	AWGN ($\sigma = 0.01$) $\rightarrow$ Poisson
Sensitivity (Sampling-rate)	Fashion-MNIST	$\{0.02, 0.05, \dots\}$	2048	2×10^{-3}	0.10	AWGN ($\sigma = 0.01$) $\rightarrow$ Poisson
Ablation (Component study)	Fashion-MNIST	0.05	512	1×10^{-3}	0.20	AWGN ($\sigma = 0.01$) $\rightarrow$ Poisson
CGI Baseline (Proxy input)	Fashion-MNIST	0.05	2048	2×10^{-3}	0.10	AWGN ($\sigma = 0.01$) $\rightarrow -$
CIFAR Robustness (8-bit)	CIFAR-10	0.05	512	1×10^{-3} (Net) 5×10^{-4} (Phys)	0.05	AWGN ($\sigma = 0.01$) $\rightarrow$ Poisson + 8-bit

Note: All random seeds and sensing matrices are fixed per experiment. Multi-run statistics in the main text report mean $\pm$ standard deviation over 5 independent runs.